

Generation Structure and Coupling Matrix from $GF(27)$:

Trace Bilinear Form, Orbit Assignment, and CKM Complexity

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Polynomial table: the polynomials for $\{5, 15, 19\}$ and $\{7, 11, 21\}$ were swapped; now $f_3 = x^3 + 2x^2 + x + 1$, $f_4 = x^3 + x^2 + 2x + 1$.
- Theorem B: note – in the Frobenius ordering of this document $M^{(34)} = -I$; the swap Gen. 2 ↔ Gen. 3 is an ordering artefact, not forced.

Status markers

- [K]** *Confirmed* — algebraically complete.
- [B]** *Established* — secured by verification script.
- [S]** *Speculative* — conjecture, not yet fully proved.

Abstract

The four irreducible polynomials $f_1, \dots, f_4$ over GF(3) define the physical orbits in GF(27)*. Their root orbits are genuine Frobenius orbits under $k \mapsto 3k \bmod 26$; the three elements correspond to the three generations. The trace bilinear form $\langle a, b \rangle = \text{Tr}_{\text{GF}(27)/\text{GF}(3)}(ab)$ gives the coupling structure between sectors: $f_3 \times f_4$ (charged leptons $\times$ quarks) is a permutation matrix **[B]**; $f_1 \times f_3$ (neutrinos $\times$ charged leptons) has normalised entries $1/\sqrt{2}$ **[B]**. Mixing angles (CKM, PMNS) remain **[S]**; the reason for the complexity of the CKM matrix is analysed.

1 Correct orbit definition

Preliminary note. Earlier documents (Docs. 339, 346, 347) used orbits $O_1 = \{1, 3, 9\}$, $O_2 = \{2, 5, 6\}$, $O_3 = \{4, 10, 12\}$, $O_4 = \{7, 8, 11\}$ as $\mathbb{Z}_{13}$ -orbits under $k \mapsto 3k \bmod 13$. These remain useful for charge and QR/NQR statements.

For the generation structure and the bilinear form, the **root orbits of the irreducible polynomials** are decisive — these are the genuine Frobenius orbits under $k \mapsto 3k \bmod 26$:

Polynomial	Root orbit	Physical role (Doc. 343)
$f_1 = x^3 + 2x + 1$	$\{1, 3, 9\}$	Neutrinos ν_1 (Majorana)
$f_2 = x^3 + 2x^2 + 1$	$\{17, 23, 25\}$	Neutrinos ν_2 (Majorana)
$f_3 = x^3 + 2x^2 + x + 1$	$\{5, 15, 19\}$	Charged leptons (e, μ, τ) / ν_3
$f_4 = x^3 + x^2 + 2x + 1$	$\{7, 11, 21\}$	Quarks (Dirac sector)

(Corrected on 1 Oct. 2026: previously $f_3 = x^3 + x^2 + 2x + 1$ for $\{5, 15, 19\}$ and $f_4 = x^3 + 2x^2 + x + 1$ for $\{7, 11, 21\}$ – in $\text{GF}(3)[x]/(x^3 + 2x + 1)$ with primitive g (order 26) the minimal polynomial of g^5 is $x^3 + 2x^2 + x + 1$, that of g^7 is $x^3 + x^2 + 2x + 1$; Docs. 342 and 343 already give this assignment. The trace values and matrices are unaffected, since they are computed with the orbit sets.)

Theorem 1.1 (Frobenius cycles of root orbits **[B]**). *The root orbits are closed under Frobenius $k \mapsto 3k \bmod 26$: $f_1 : 1 \rightarrow 3 \rightarrow 9 \rightarrow 1$; $f_2 : 17 \rightarrow 25 \rightarrow 23 \rightarrow 17$; $f_3 : 5 \rightarrow 15 \rightarrow 19 \rightarrow 5$; $f_4 : 7 \rightarrow 21 \rightarrow 11 \rightarrow 7$. The three elements of each orbit correspond to the three generations.*

2 Trace bilinear form

$$\langle g^k, g^j \rangle = \text{Tr}_{\text{GF}(27)/\text{GF}(3)}(g^{k+j}) = g^{k+j} + g^{3(k+j)} + g^{9(k+j)} \in \text{GF}(3).$$

Values in $\{0, 1, 2\}$; as real numbers: $0 \mapsto 0$, $1 \mapsto +1$, $2 \mapsto -1$.

3 Theorem B — Coupling matrix $f_3 \times f_4$ (CKM sector)

Theorem 3.1 (Permutation matrix in the quark-lepton sector **[B]**). *The coupling matrix between charged leptons (f_3) and quarks (f_4):*

$$M^{(34)} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

This is a permutation matrix: lepton Gen. 1 couples to quark Gen. 1; lepton Gen. 2 to quark Gen. 3; lepton Gen. 3 to quark Gen. 2. The swap Gen. 2 $\leftrightarrow$ Gen. 3 is algebraically forced **[B]**. (Note of 1 Oct. 2026: in the Frobenius ordering fixed by this document ($f_3: 5 \rightarrow 15 \rightarrow 19$, $f_4: 7 \rightarrow 21 \rightarrow 11$, generation = position in the cycle) the trace form gives $M^{(34)} = -I$, i.e. diagonal: Gen. i couples to Gen. i . The printed matrix with the swap 2 $\leftrightarrow$ 3 arises only when f_4 is read in the sorted order $\{7, 11, 21\}$ of the table; it is an ordering artefact and not algebraically forced.)

The experimental CKM matrix is nearly diagonal with small mixing angles. The Galois coupling matrix gives the zero structure (which Gen. combinations do not couple), but not the mixing angles themselves. **[S]**

4 Theorem C — Coupling matrix $f_1 \times f_3$ (PMNS sector)

Theorem 4.1 (Equal-weight coupling in the neutrino sector **[B]**).

$$M^{(13)} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad |M_{ij}^{(13)}|_{\text{norm}} = \frac{1}{\sqrt{2}} \text{ for all non-zero entries.}$$

Each neutrino generation couples to two charged lepton generations with weight $1/\sqrt{2}$ — consistent with the large PMNS mixing angles ($\theta_{23} \approx 49^\circ$, $\sin^2 \theta_{23} \approx 0.57$, near $(1/\sqrt{2})^2 = 0.5$, NuFIT 5.3 [5]; maximal mixing would be 45°). **[S]**

5 Why CKM mixing angles are complex

The CKM matrix has four independent parameters (Wolfenstein): $\lambda \approx 0.225$, $A \approx 0.811$, $\rho \approx 0.124$, $\eta \approx 0.356$. The CP-violating phase $\delta_{CP} \approx 68^\circ$ ($e^{i\delta_{CP}} = 0.375 + 0.927i$) makes the matrix essentially complex (CP violation in kaon and B-meson decays, observed 1964 and 2001).

What $\text{GF}(27)^*$ provides and what it does not. $\text{Gal}(\text{GF}(27)/\text{GF}(3)) = \mathbb{Z}_3$ has three irreducible representations over $\mathbb{C}$: trivial (1), $\omega = e^{2\pi i/3}$ (120°), ω^2 (240°). If the coupling matrix transforms under ρ_1 , a complex phase arises — the most plausible Galois candidate for the CP sector. **Problem:** ω gives 120° , not 68° (gap 52°). No algebraically forced candidate **[S]**.

Three steps needed for full CKM from $\text{GF}(27)$: (1) **Real zero structure:** permutation matrix from trace bilinear form **[B]**. (2) **Mixing angles:** corrections to permutation matrix giving $\lambda \approx 0.225$, $A \approx 0.811$. Three checked paths: direct ξ -corrections ($\sin \theta_C / \xi \approx 1685$, too large); $\xi^{|k_i - k_j|}$ weighting ($\sim 10^{-8}$, too small); Galois quotients $2/9 \approx 0.222$ and $3/13 \approx 0.231$ (near, but not forced). No candidate **[S]**. (3) **CP phase:** complex embedding of Galois group forcing $\delta_{CP} \approx 68^\circ$. $\mathbb{Z}_3$ phases give 120° , not 68° . $\text{GF}(27)^*$ has no natural complex structure; the trace bilinear form gives values in $\{0, 1, 2\} \subset \text{GF}(3)$ **[S]**.

Important distinction: not derivable $\neq$ not usable. $\delta_{CP} \approx 68^\circ$ can be set as a phase anchor — structurally identical to $v = 246$ GeV or C_{conv} : empirically given, not freely adjustable, and once set all CP-violating observables are fully determined. Deriving δ_{CP} from first principles remains **[S]**; computing with it is immediately possible.

6 Summary

Table 1: Results Doc. 348

Theorem	Content	Status
A	Root orbits = correct Frobenius orbits; 3 elements = 3 generations	[B]
B	$f_3 \times f_4$ coupling matrix = permutation matrix	[B]
C	$f_1 \times f_3$ coupling matrix with $1/\sqrt{2}$ weights	[B]
D	Full CKM/PMNS angles from bilinear form	[S]

Verification script

pruef_348_ckm.py — Theorems A–C, all assertions passed **[B]**.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations and conclusions by Johann Pascher (ORCID 0009-0000-6518-4064).

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